



control units

mindy TT0

Instructions and warnings for the fitter

Istruzioni ed avvertenze per l'installatore

Instructions et recommandations pour l'installateur

Anweisungen und Hinweise für den Installateur

Instrucciones y advertencias para el instalador

Instrukcje i uwagi dla instalatora

Warnings

The mindy TT0 control unit has been designed to control asynchronous single-phase electric motors for automating awnings, shutters and the like. It must be installed by qualified fitters in total compliance with cur-

rent electrical and safety standards. During installation, particular attention must be paid to ensure the casing is protected to the "IP" level required by the application.

1) Description of the product

The mindy TT0 control unit controls asynchronous single-phase electric motors with "COMMON" "OPEN" "CLOSE" connections and can be used for automating awnings, shutters, skylights and similar applications.

The control unit incorporates a radio receiver working at a frequency of 433.92 MHz using rolling code technology which ensures elevated levels of safety. It is possible to store the codes of up to 14 radio commands of the "ERGO" and "PLANO" series (figs. 1.a and 1.b) or radio-sensors for each control unit. After each command, the

motor is powered for about 2 minutes and an electrical limit switch fitted to the motor or the automation unit stops movement when the required position has been reached. Additional functions can be programmed by the radio control units, a "beep" guides users through the various phases. As well as using the radio control, the control units can also be operated with an external button (with a step-by-step function) or via Bus ("TTBUS"). Optional wind, sun and rain sensors automatically manage the system according to weather conditions.

2) Installation

⚠ WARNING: The electrical systems and automation units must be installed by qualified technicians in full compliance with the laws in force. All connections must be made the unit disconnected from the power supply.

To install and fix the casing, please refer to figure 2.

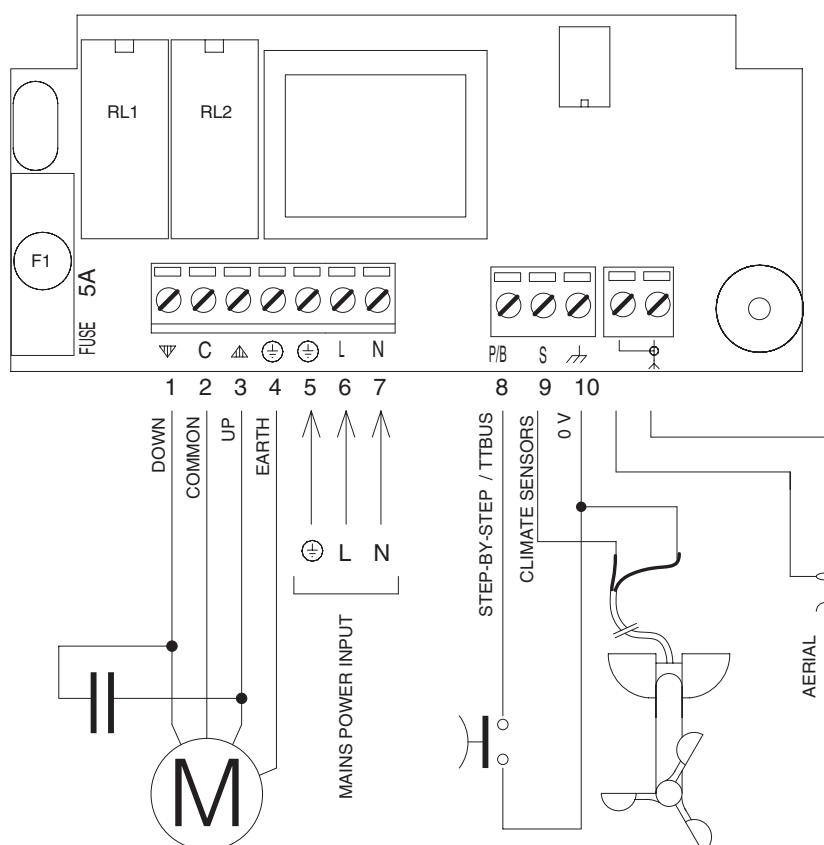
When making holes in the casing to pass the wires through, take suitable precautions to ensure the required IP protection level is maintained.

The wires must always be passed through the bottom of the casing using the supplied core hitch cut to a suitable diameter (see fig. 2).

2.1) Wiring diagram

⚠ Strictly keep to the wiring diagram; if you have any doubts do NOT make experiments but consult the relative technical sheets which are also available on the web site "www.niceforyou.com".

An incorrect connection may cause serious damage to the control unit.



2.1.1) Connecting the motor

The asynchronous single-phase electric motor must be connected between terminals 1-2-3; n° 2 is the "common".

"DOWN" corresponds to the ▼ button of the remote control units, "UP" to the ▲ button (direction of anemometer input). If the sense of rotation is incorrect, swap the connections of terminals 1 and 3 or invert the software with the remote control unit (see Table "A3")

⚠ WARNING: Do not connect more than one motor for each control unit

2.1.2) Power input

The main power input of the control unit (earth, phase, neutral) must be connected using terminals 5-6-7 as shown in figure 3 "Wiring diagram"

2.1.3) "Step-by-Step" input:

To control the automation unit manually a simple button can be connected (terminals 8-10). The operating sequence is the following: up-stop-down-stop.

If the button is kept pressed for more than 3 seconds (but less than 10) an up manoeuvre is always carried out (corresponding to the ▲ button of the radio control units).

If the button is kept pressed for more than 10 seconds a down manoeuvre is always carried out (corresponding to the ▼ button). This function can be used to "synchronise" more than one motor to carry out the same manoeuvre, regardless of its operating status.

2.1.4) "TTBUS" input:

"TTBUS" is a Bus that was developed to manage control units for awnings and shutters and motors fitted with control units. The Bus can individually control up to 100 units by simply connect-

ing them in parallel with just 2 wires (terminals 8-10). Further information is contained in the remote controls via "TTBUS" instructions.

2.1.5) Weather sensors:

A simple wind sensor (anemometer) or a special wind-sun-rain sensor can be connected to "Weather sensors" input (terminals 9-10). Up to 5 control units can be connected in parallel to the same sensor respecting the polarity of the signals (on all the control units, terminal 9 must be connected to 9 and terminal 10 to 10).

⚠ WARNING: the anemometer gives a command input equivalent to the ▲ button on the remote control units

3) Programming

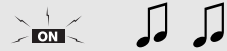
Each radio control unit is recognised by the receiver incorporated in control unit by means of a unequivocal "code". A "storing" phase must therefore be performed in order to allow the control unit to recognise each single radio control unit.

⚠ ATTENTION: All the storing sequences are timed, that is, they must be completed within the programmed time limits.

• **For radio control units with more than one "unit", choose the unit to associate the control unit with before proceeding with the storing phase.**

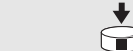
• **Programming via radio may be done on all the control units within the range of the transmitter; only the one involved in the operation should be kept switched on, therefore.**

When the memory contains no codes the first radio control unit can be entered as follows:

Table "A1"	Memorising the first transmitter (fig. 4)	Example
1.	As soon as the control unit is powered, 2 long beeps will sound	
2.	Within 5 seconds press and hold down button  of the transmitter to memorise (for approx. 3 seconds)	 3s
3.	Release button  when you hear the first of the 3 beeps confirming memorisation	

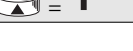
N.B.: If the control unit already contains codes, 2 short beeps will be heard when it is switched on. This means that the above procedure is not valid and another memorisation procedure must be used.

When one or more transmitters have already been memorised, others may be enabled as follows:

Table "A2"	Memorising other transmitters (fig. 5)	Example
1.	Press and hold down button  of the new transmitter until you hear a beep (after about 5 seconds)	 5s
2.	Press button  of a previously enabled transmitter slowly 3 times (old)	 X3
3.	Press button  of the new transmitter again.	
4.	At the end, 3 beeps will indicate that the new transmitter has been memorised correctly.	

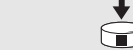
N.B.: If the memory is full (14 codes), 6 beeps will indicate that the transmitter can no longer be memorised.

When the direction of movement with respect to the radio control unit buttons must be inverted, proceed as follows

Table "A3"	Inverting the direction of the motor with respect to the controls (fig. 6)	Example
1.	Press and hold down button  of a previously memorised transmitter until you hear a beep (after about 5 seconds)	 5s
2.	Then hold down both the  and the  button (approx. 4 seconds) until you hear the first of the 3 beeps confirming that the direction has been inverted	 4s
3.	Test the new direction of the manoeuvre,  should correspond to "up" while  should correspond to "down".	 =   = 

⚠ if the anemometer triggers, this will cause the motor to carry out a manoeuvre equivalent to the  button

If a wind sensor is connected to the "sensors" input it is possible to select the cut-in level from 3 possible levels: 1= 15 km/h, 2= 30 km/h and 3= 45 km/h (the level was originally n° 2). When the level is exceeded for over 3 seconds, a command equivalent to the  button is activated and all other movements are blocked until the wind returns to under the programmed level. To modify the programmed level:

Table "A4"	Changing the "wind" protection cut-in level (fig. 7)	Example
1.	Press button  of a previously memorised transmitter until you hear a beep (after about 5 seconds)	 5s
2.	Slowly press the  button a number of times (1, 2 or 3) equal to the required level	 X1= 15 km/h X2= 30 km/h X3= 45 km/h
3.	After a few moments you will hear a number of beeps equal to the required level	 X1= 15 km/h X2= 30 km/h X3= 45 km/h
4.	Press button  to confirm, otherwise wait for at least 5 seconds without confirming in order to abort the procedure without changing the previous level	

If you need to delete all the data contained in the memory of the control unit, carry out the following procedure.

The memory can be deleted:

- with a non-memorised transmitter starting from point A.
- with a previously memorised transmitter starting the procedure from point N° 1

The following can be deleted:

- only the transmitter codes, finishing at point N° 4
- all data (transmitter codes, directions, wind levels, TTBUS addresses, etc.), completing the procedure.

Table "A5"	Memory deletion (fig. 8)	Example
➔ A	Switch the motor off, activate the Step-by-Step input and keep it active until the end of the procedure	 
B	Power the motor and wait for the 2 initial beeps	  
➔ 1	Press and hold down button  of a transmitter until you hear a beep (after about 5 seconds)	  5s
2	Hold down the  button of the transmitter until you hear 3 beeps; release the  button exactly <u>during the third beep</u>	    
3	Hold down the  button of the transmitter until you hear 3 beeps; release the  button exactly <u>during the third beep</u>	    
➔ 4	Hold down the  button of the transmitter until you hear 3 beeps; release the  button exactly <u>during the third beep</u>	    
5	If you wish to delete all the data, press the  and  buttons together within 2 seconds and then release them.	within 2 s  

After a few seconds 5 beeps signal that all the memorised codes have been deleted.

4) What to do if... a small troubleshooting guide!

When the board powered, the 2 beeps will not be heard and the Step-by-Step input does not control any movement.

Check the control unit is correctly powered: mains voltage must be present between terminals 6-7, a voltage of about 24Vdc must be present between terminals 8-10 with the Step-by-Step button disconnected. Check the 5A fuse F1 if necessary.

After a radio command, 6 beeps are heard and the manoeuvre does not start

The radio control unit is out of sync, repeat the transmitter code storing procedure.

After a command, 10 Beeps sound and then the manoeuvre begins

The self-diagnosis of the stored parameters has revealed a fault (TTBUS addresses, wind speed and sun power, directions are incorrect). Check and repeat programming if necessary

5) Technical specifications

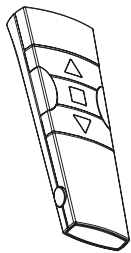
Electronic control unit

Power input	: 230Vac 50/60Hz or 120Vac 50/60 Hz depending on the versions (see value on label)
Maximum power of motors	: 600W for 230Vac version; 400W for 120Vac version
Signal voltage (step-by-step, sensors)	: Approx. 24Vdc
Working temperature	: -10 ÷ 70 °C
Dimensions / weight	: 120 x 80 x 50 300 g
Protected to "IP"	: 55 (casing undamaged)
Wind sensor (anemometer) levels	: Approx. 30 or 15 or 45 km/h (with anemometer at 0.4 Hz per km/h)
Length of signal cables (step-by-step, sensors)	: max. 30m if near other cables, otherwise 100m

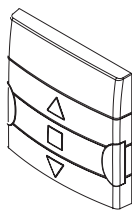
Radio receiver

Frequency	: 433.92 MHz
Coding	: 52 Bit rolling code FLOR
Range of ERGO and PLANO transmitters	: Estimated 200 m in the open air without obstacles and 35 m inside buildings.

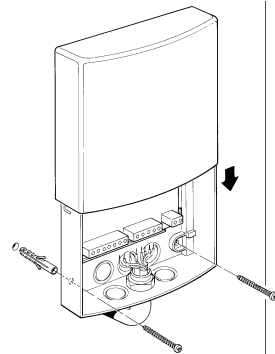
Nice S.p.a. reserves the right to modify its products without any notice



1a ERGO



1b PLANO



2

4



~3 s



5

+5 s



X3



X1



6

+5 s

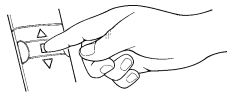


~4 s



7

+5 s



X1= 15 Km/h
X2= 30 Km/h
X3= 45 Km/h



8

+5 s



3° bip



3° bip



3° bip



Only TX

2 s

All

-2 s X 1



Dichiarazione di conformità / Declaration of conformity

N°: mindy TT0 Rev 0

Nice S.p.a. via Pezza Alta, 13 Rustignè Oderzo (TV) ITALY

dichiara che il prodotto: "mindy TT0" / declares that the product: "mindy TT0"

centrale di comando per tapparelle; tende da sole e serrande / control unit for rolling shutters and awnings

risulta conforme ai requisiti essenziali di sicurezza delle direttive: / complies with the essential safety requirements of the following directives:

73/23/CEE Direttiva Bassa Tensione / Low Voltage Directive

89/336/CEE Direttiva compatibilità elettromagnetica / Electromagnetic Compatibility Directive

1995/5/CE Apparecchiature radio e terminali di telecomunicazione / Radio equipment and telecommunications terminals

Risulta conforme a quanto previsto dalle altre norme e/o specifiche tecniche di prodotto specifications / complies with other product standards and/or technical specifications

LVD: CEI EN 60335-1/1995 + A12 + A1 + A13 + A14

EMC: EN 55014; EN 61000-3-2; EN50082-1

R&TTE: ETS 300683; EN 300220-3; EN 60950

Data /date

19 Settembre 2001

Amministratore Delegato / General Manager
Lauro Buoro



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